

Mathematics Standard 1

HSC marking feedback 2025

General feedback

Students should:

- show relevant mathematical reasoning and/or calculations
- read the question carefully to ensure they do not miss important components of the question
- have a clear understanding of the intent of the question and key words in the question such as show, solve, evaluate, hence, calculate
- use the Reference Sheet where appropriate
- ensure the solution is legible and follows a clear sequence
- engage with any stimulus material provided and refer to it in their response when required by the question
- check their solution answers the question
- round off numerical solutions only at the final step of the solution
- construct graphs neatly, with precision, and display all relevant information as required by the question
- interpret information presented in graphs across a range of contexts
- understand when to use relevant calculator functions
- note carefully any information in the questions which supplies units of measurement.

Section II

Question 11

In better responses, students were able to:

- identify the two shapes, a rectangle and a triangle, and recognise the figure as a composite shape
- split the composite figure into the rectangle and triangle and apply the correct formula to each
- identify the triangle's base length correctly as 6 cm by using diagram measurements
- find or infer missing side lengths from the diagram.

Areas for students to improve include:

- calculating areas correctly and distinguishing area from perimeter
- recognising and splitting composite shapes into standard parts
- finding missing or inferred side lengths from given measures.

Question 12

In better responses, students were able to:

- find the increase in heart rate
- calculate the correct percentage increase.

Areas for students to improve include:

- calculating a percentage correctly.

Question 13

In better responses, students were able to:

- recognise the perpendicular height of a trapezium and identify the two parallel sides
- show a clear substitution into their formula from part (a)
- use the expression from part (a) to solve for the unknown with the given area
- identify the given area as the subject of the equation and solve.

Areas for students to improve include:

- identifying the correct use of the perpendicular height for the area of a trapezium formula
- distinguishing between area and dimensions of a shape
- recognising the appropriate units for length and area.

Question 14

In better responses, students were able to:

- identify the shortest path and find the minimum travel time
- explain if the path in part (a) is still the shortest path from A to F after the new roads are added.

Areas for students to improve include:

- identifying the shortest path and ensuring vertices are not missed
- ensuring that they check different paths, as the least number of edges is not always the shortest path
- comparing the new shortest path to the original shortest path and stating whether the original path is shorter.

Question 15

In better responses, students were able to:

- connect the dependent variable with the y -axis
- recognise that the gradient was negative
- plot points correctly
- use the points given to find the gradient
- recognise that the line would go into negative values for the average minutes per day spent exercising.

Areas for students to improve include:

- posing a question, rather than providing a statement
- using bivariate datasets to retrieve information about form and direction
- demonstrating understanding of why negative time is not possible
- demonstrating understanding of the difference between extrapolation and interpolation.

Question 16

In better responses, students were able to:

- write a direct variation statement and calculate the constant of proportionality
- use the constant of variation to find the new mass
- divide the mass of the box by the area of its base to find the constant of variation
- use the unitary method to find the mass of the box
- link the variation value to the new area.

Areas for students to improve include:

- demonstrating understanding of what is meant by direct variation
- finding and using the constant of variation
- writing variation equations using k as the variation constant.

Question 17

In better responses, students were able to:

- calculate the cost of buying 80 litres of paint in 10-litre cans correctly
- calculate the cost of buying 80 litres of paint in 4-litre cans correctly
- calculate the savings by finding the difference between the cost of 10-litre and 4-litre cans.

Areas for students to improve include:

- showing the difference of two calculated values
- calculating the amount saved for the whole amount of 80 litres, not just 1 litre.

Question 18

In better responses, students were able to:

- identify that 1.25 hours is equivalent to 1 hour and 15 minutes
- calculate addition of time values
- use the given distance and time to find the speed.

Areas for students to improve include:

- converting hours to minutes
- demonstrating understanding that 1.25 hours does not equal 1 hour and 25 minutes.

Question 19

In better responses, students were able to:

- identify the correct row from the tax table
- find the difference between taxable income and the final number in the correct row
- multiply the found difference by 0.37
- calculate the Medicare levy payable by multiplying their taxable income by 2%.

Areas for students to improve include:

- demonstrating understanding that 37 cents is the same as 0.37
- using brackets or a separate line of working to calculate $(148600 - 135000)$
- selecting the correct row from the tax table and using it to calculate tax payable
- demonstrating understanding that the Medicare levy is calculated on the taxable income, not the tax payable.

Question 20

In better responses, students were able to:

- show the correct use of scale
- connect the scale in centimetres or grid boxes to distance in real life
- count the number of squares and convert to the correct length
- apply Pythagoras' theorem correctly and clearly, including calculating the square root
- round to a specific number of significant figures
- identify and use directions to find bearings.

Areas for students to improve include:

- connecting the scale to either centimetres or grid boxes
- calculating actual lengths on scale diagrams through a measurement and then applying the scale
- converting between units of length
- demonstrating understanding of when to square and when to calculate the square root
- demonstrating understanding of the difference between rounding to significant figures and rounding to decimal places
- demonstrating understanding of the difference between true and compass bearings.

Question 21

In better responses, students were able to:

- calculate the number of days correctly
- use the compound interest formula correctly to calculate the amount owed.

Areas for students to improve include:

- calculating the interest per day as a percentage or decimal
- converting the interest rate and time periods to the correct unit.

Question 22

In better responses, students were able to:

- find the base length of each right-angled triangle is 2.4
- find the height of the triangle using Pythagoras' theorem correctly
- recognise that the height of an isosceles triangle bisects the base of the triangle.

Areas for students to improve include:

- demonstrating understanding of the difference between calculating the length of the hypotenuse and the length of one of the shorter sides
- expressing a quantity as a percentage of another quantity.

Question 23

In better responses, students were able to:

- provide two equations correctly using the pronumerals given in the question
- interpret the information to retrieve correct pronumerals.

Areas for students to improve include:

- demonstrating understanding of the need to write two equations
- demonstrating understanding of the format of linear equations.

Question 24

In better responses, students were able to:

- calculate the total stamp duty
- calculate the interest on the total borrowed, not just on the cost of the car
- find the total owing correctly, and understand it was the total borrowed, plus the interest.

Areas for students to improve include:

- calculating the stamp duty on the sale price of the car
- demonstrating understanding of what is meant by a monthly repayment.

Question 25

In better responses, students were able to:

- compare future values of option A and option B from the respective compound interest and simple interest calculations
- calculate compound interest, remembering to change the value for 'r' and 'n' when compounding periods are not in years
- recognise that the future value from compound interest is inclusive of investment.

Areas for students to improve include:

- converting the rate and periods into the compounding periods stated in the question, particularly for interest compounding monthly
- demonstrating understanding that interest or future values should be compared to see the difference in interest earned with two different interest types.

Question 26

In better responses, students were able to:

- use the given scale/ratio to convert diagram lengths to real lengths
- calculate the total cost components, that is, tile cost and labour
- apply the 15% extra correctly by multiplying by 1.15 or adding 15% to the original area, to allow for cutting and breakage
- convert units when necessary and use ratios/scale consistently.

Areas for students to improve include:

- applying the scale factor and ratios accurately when converting lengths
- computing area reliably and not relying on counting grid squares
- increasing amounts by 15% correctly.

Question 27

In better responses, students were able to:

- use the formula to find the salvage values at the end of the 9th and 10th years
- calculate the amount the car depreciates in the 10th year
- demonstrate understanding that the 10th year is not at the end of year number 10, but how much the car has dropped in value during that 10th year.

Areas for students to improve include:

- demonstrating understanding that depreciating means the value is decreasing as time progresses
- demonstrating understanding that with declining balance, the depreciating amount is not a constant value but rather a rate/percentage
- working out a rate using a graph.

Question 28

In better responses, students were able to:

- convert units correctly between kilograms and grams
- use the unitary method to find the mass of one coin, that is, divide the total mass by number of coins, and scale up, that is, multiply by 500, to get the mass of 500 coins
- use ratios/rates confidently, write correct units, show clear steps, and treat separate parts of the question independently.

Areas for students to improve include:

- converting between grams and kilograms
- applying the unitary method consistently and show working for each step.